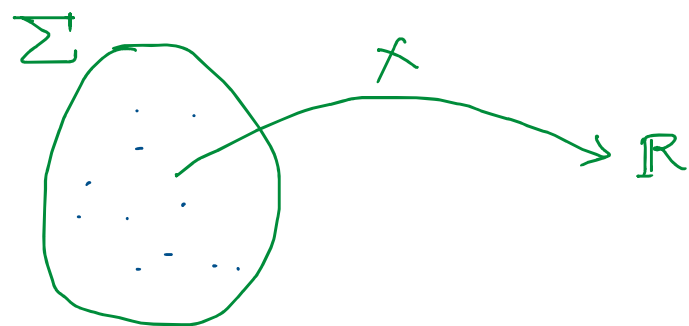


Random process or experiment

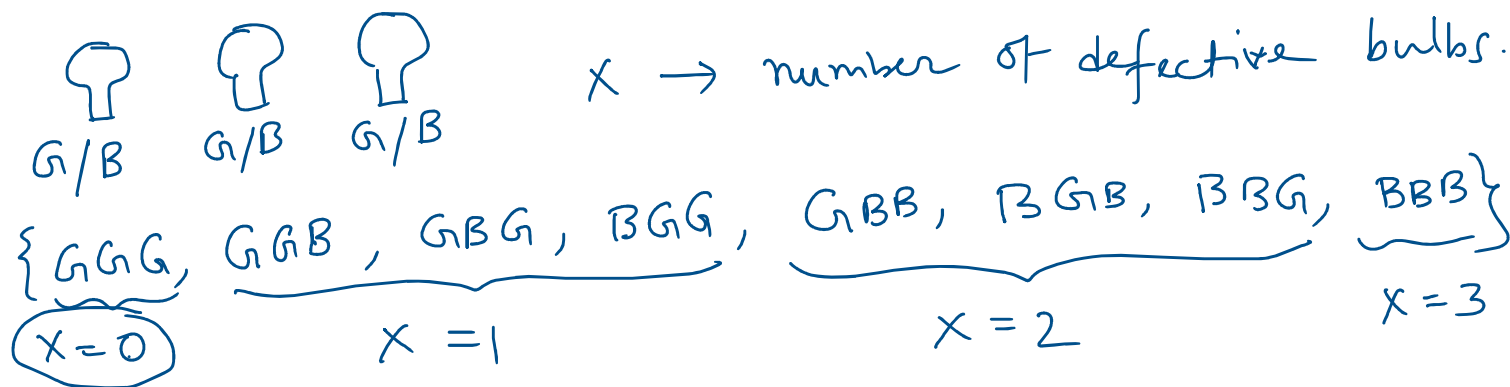
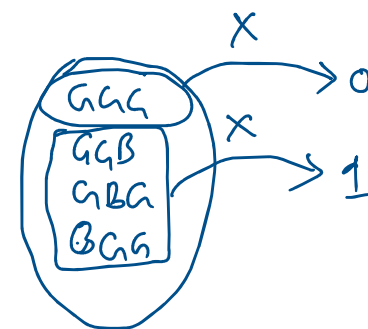
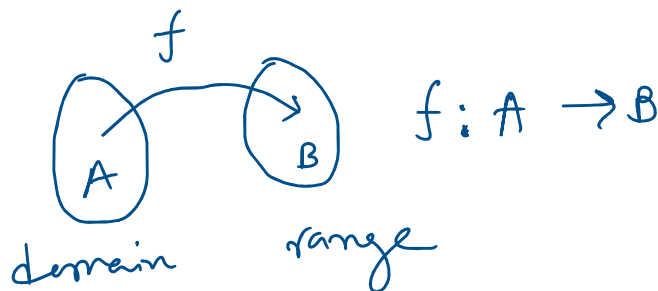


Sample space
(Set of all possible outcomes)

X is a function which maps the sample space to real numbers.

$$X: \Sigma \rightarrow \mathbb{R}$$

Random variable



$$P(X=0) = \frac{1}{8}; p_X(0)$$

$$P(X=1) = \frac{3}{8}; p_X(1)$$

$$P(X=2) = \frac{3}{8}$$

$$P(X=3) = \frac{1}{8}$$

$$P(A \cap B) = P(A) \cdot P(B) \rightarrow \text{independent}$$

$$P(A \cap B) = P(\emptyset) = 0 \rightarrow \text{disjoint}$$

Biased coin: $p_H = 0.6$, $p_T = 0.4$

3 coin toss

$X = \# \text{ heads}$

{TTT}

{HTT, THT, TTH}

$X = 0$

$X = 1$

$X = 2$

$X = 3$

$$P(X=0) = (0.4)^3$$

$$P(X=1) = 3 \cdot (0.6) \cdot (0.4)^2$$

$$P(X=2) = 3 \cdot (0.6)^2 \cdot (0.4)$$

$$P(X=3) = (0.6)^3$$

$$P(X=x) = \frac{k}{2^x} \quad x=1,2,3$$

$$= 0 \quad \text{otherwise}$$

$$P(X=0) = 0$$

$$P(X=1) = \frac{k}{2}$$

$$P(X=2) = \frac{k}{2^2}$$

$$P(X=3) = \frac{k}{2^3}$$

$$P(X=4) = P(X=5) = 0$$

$$\frac{k}{2} + \frac{k}{2^2} + \frac{k}{2^3} = 1$$

$$\Rightarrow k =$$

$$\frac{X=0}{p_x(0)}, \frac{X=1}{p_x(1)}, \frac{X=2}{p_x(2)}, \frac{X=3}{p_x(3)}, \frac{X=4}{p_x(4)}$$

$$P(X=x)$$

$$F_x(0) = p_x(0)$$

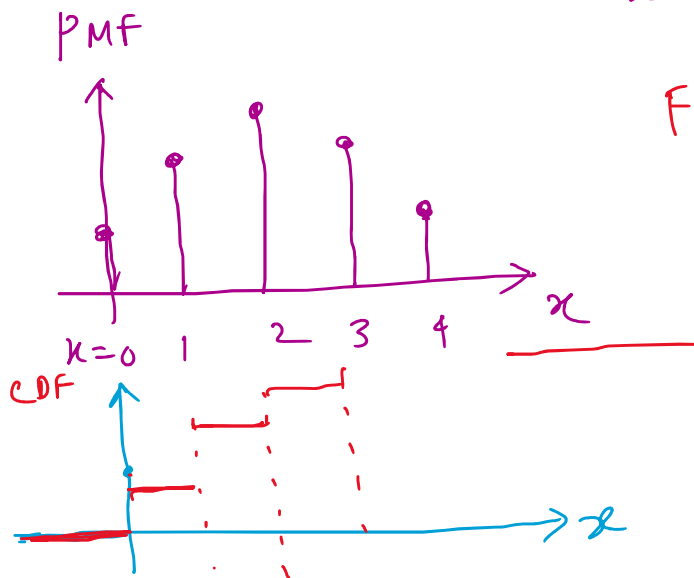
$$F_x(1) = P(X \leq 1) = P(X=0) + P(X=1) = p_x(0) + p_x(1)$$

$$F_x(2) = p_x(0) + p_x(1) + p_x(2)$$

$$F_x(3) = \sum_{x=0}^3 p_x(x)$$

$$F_x(x) = P(X \leq x)$$

$$F_x(5) = P(X \leq 5) = 1$$



PMF
 $P(X=0) = 0$

$$P(X=1) = \frac{4}{7}$$

$$P(X=2) = \frac{2}{7}$$

$$P(X=3) = \frac{1}{7}$$

$$P(X=4) = 0$$

$$\vdots$$

$F_X(x)$

$$F_X(1.5) = F_X(1)$$

CDF
 $F_X(0) = 0$

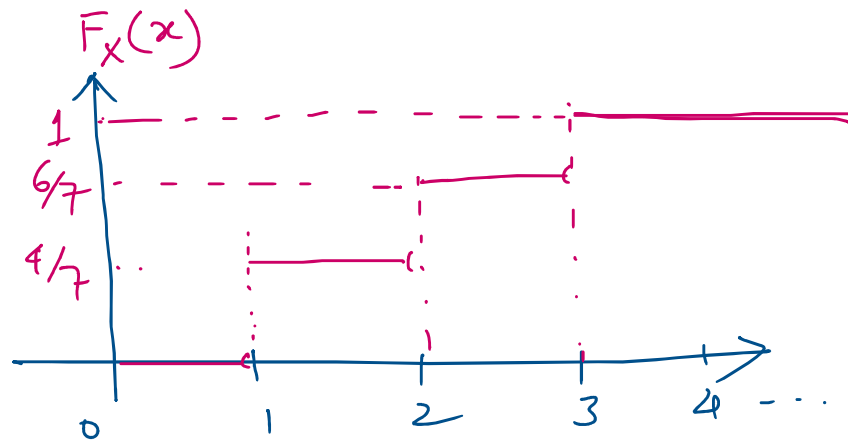
$$F_X(1) = \frac{4}{7}$$

$$F_X(2) = \frac{6}{7}$$

$$F_X(3) = 1$$

$$F_X(4) = 1$$

$$\vdots$$



$$F_X(0.99999 \dots 9) = 0$$

$$F_X(1) = \frac{4}{7}$$

$$F_X(1.99999 \dots 9) = \frac{4}{7}$$

$$F_X(2) = \frac{6}{7}$$

$$\begin{matrix} (w_1) & (w_2) & (w_3) & \dots & (w_n) \\ x_1 & x_2 & x_3 & \dots & x_n \end{matrix}$$

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\bar{x}_{\text{weighted}} = \frac{x_1 w_1 + x_2 w_2 + \dots + x_n w_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum_{i=1}^n x_i w_i}{\sum_{i=1}^n w_i}$$

Say $\boxed{\sum_{i=1}^n w_i = 1}$ then $\bar{x}_{\text{weighted}} = \sum_{i=1}^n x_i w_i$

pmf $\sum_x p_x(x) = 1$

$$E[x] = \sum_x x p_x(x)$$

3 servers. : 50 % chance for each server to receive high priority req.
 $X = \#$ high priority requests.

x	$p_x(x)$
$X = 0$	$p_x(0) = \frac{1}{8}$
$X = 1$	$p_x(1) = \frac{3}{8}$
$X = 2$	$p_x(2) = \frac{3}{8}$
$X = 3$	$p_x(3) = \frac{1}{8}$

$$\begin{aligned} E[X] &= 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} \\ (\mu) &= \frac{12}{8} = 1.5 = \mu \end{aligned}$$

$$\begin{array}{c} (X - \mu) \\ \hline -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{array}$$

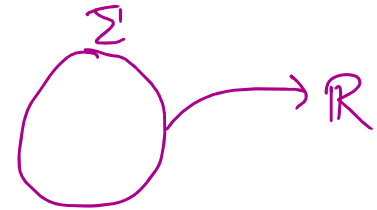
$$\text{Var}(X) = \sum_x (x - \mu)^2 p_x(x) =$$

$$\begin{aligned} &= (-1.5)^2 \frac{1}{8} + (-0.5)^2 \times \frac{3}{8} + (0.5)^2 \frac{3}{8} + (1.5)^2 \frac{1}{8} \\ &= 2 \times \left(\frac{3}{2}\right)^2 \times \frac{1}{8} + 2 \left(\frac{1}{2}\right)^2 \cdot \frac{3}{8} = 2 \left[\frac{9}{32} + \frac{3}{32} \right] = \frac{24}{32} \\ &= 0.75 \end{aligned}$$

x	x^2	$p_x(x)$
0	0	$\frac{1}{8}$
1	1	$\frac{3}{8}$
$x = \boxed{2} \rightarrow \boxed{4} \rightarrow \textcircled{\frac{3}{8}}$	3	$\frac{1}{8}$

$$\begin{aligned}\mathbb{E}[x^2] &= 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 4 \times \frac{3}{8} + 9 \times \frac{1}{8} \\ &= \frac{24}{8} = \underline{3}\end{aligned}$$

$$\mathbb{E}[x] = \frac{3}{2}$$

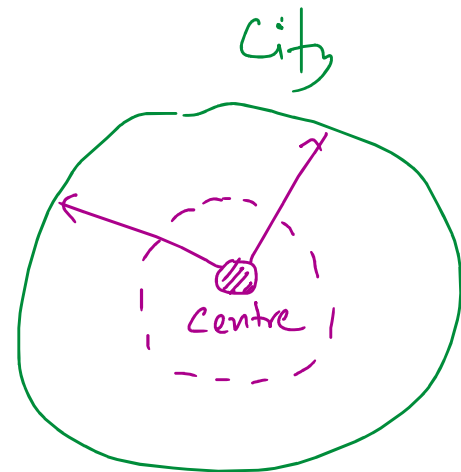
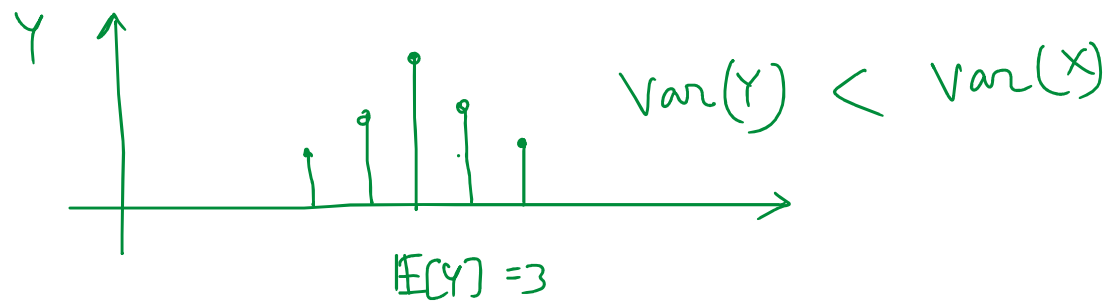
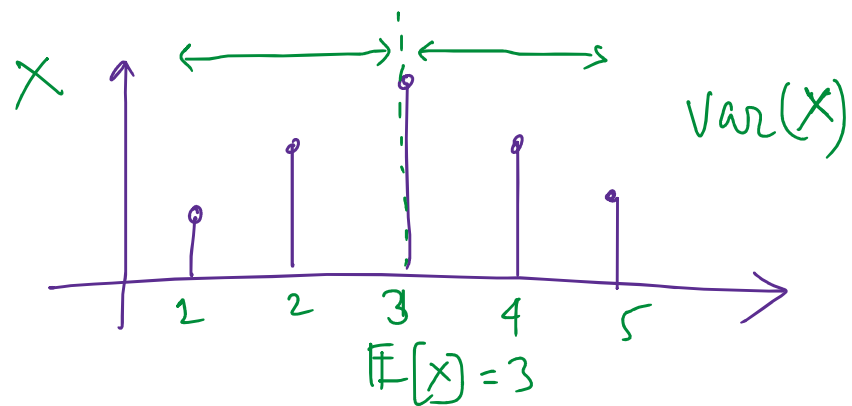


$$\begin{aligned}\text{Var}(x) &= \mathbb{E}[x^2] - (\mathbb{E}[x])^2 = 3 - \left(\frac{3}{2}\right)^2 \\ &= 3 - \frac{9}{4} = \frac{12 - 9}{4} = \frac{3}{4} = 0.75\end{aligned}$$

$$\text{Var}(x) = \mathbb{E}[(x - \mathbb{E}[x])^2] = \mathbb{E}[x^2] - (\mathbb{E}[x])^2$$

$Y = f(X)$ then $E[Y] = ?$

$$E[Y] = \sum_x f(x) \cdot p_x(x)$$



$n \rightarrow$ trials

$p \rightarrow$ prob of success of a single trial, $(1-p) \rightarrow$ probability of failure

$$\underbrace{P(X=k)} = {}^nC_k \cdot p^k \cdot (1-p)^{n-k}$$

probability of k success out of n trials.
 $0 \leq k \leq n$

5 coin toss, $p = 0.7$

$$P(X=2) = {}^5C_2 \cdot (0.7)^2 \cdot (0.3)^3$$

$$\sum_{k=0}^n P(X=k) = 1$$

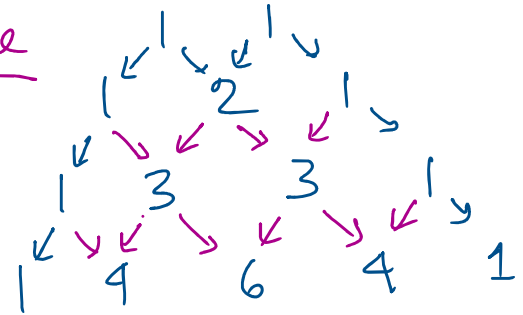
$$\sum_{k=0}^n {}^nC_k p^k \cdot (1-p)^{n-k} = 1$$

$$(1+x)^2 = 1 + 2x + x^2$$

$$(1+x)^3 = 1 + 3x + 3x^2 + x^3$$

$$(1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$$

Pascal's triangle



$$(1+x)^n = {}^nC_0 \cdot 1 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_n x^n$$

$$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} \cdot b + {}^nC_2 a^{n-2} b^2 + {}^nC_3 a^{n-3} \cdot b^3 + \dots + {}^nC_n b^n$$

$$\underbrace{p \quad 1-p}_{1} = \sum_{k=0}^n {}^nC_k p^k (1-p)^{n-k}$$

PMF for Poisson's distribution

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$x = 0, 1, 2, \dots, n, \dots$$

$\lambda \Rightarrow$ parameter

$$\mathbb{E}[X] = \lambda \quad \& \quad \text{Var}[X] = \lambda$$

$$\mathbb{E}[X] = \sum_x x p_x(x) = \sum_x x \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$\begin{aligned} x! &= x(x-1)(x-2)\dots 1 \\ &= x(x-1)! \end{aligned}$$

$$\begin{aligned} &= \sum_x \frac{(e^{-\lambda} \cdot \lambda^x) \cdot x}{x(x-1)!} = \sum_x \frac{e^{-\lambda} \cdot \lambda^x \cdot x}{x(x-1)!} \\ &= e^{-\lambda} \cdot \lambda \cdot \sum_x \frac{\lambda^{x-1}}{(x-1)!} \\ &= e^{-\lambda} \cdot \lambda \cdot e^{\lambda} = \lambda \end{aligned}$$

Q2: A call center receives an average of $\lambda = 3$ calls per minute (Poisson). What is the probability of receiving exactly 0 calls in a minute?

$$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!} \quad (\lambda = 3)$$

$$P(X = 0) = \frac{e^{-3} \cdot 3^0}{0!} = e^{-3}$$